

New methodology for calculating damage variables evolution in Plastic Damage Model for RC structures



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ABSTRACT

The behavior of reinforced concrete (RC) structures under severe demands, as strong ground motions, is highly complex; this is mainly due to joint operation of concrete and steel, with several coupled failure modes. Furthermore, given the increasing awareness and concern for the important seismic worldwide risk, new developments have arisen in earthquake engineering. Nonetheless, simplified numerical models are widely used (given their moderate computational cost), and many developments rely mainly on them. The authors have started a long-term research whose final objective is to provide, by using advanced numerical models, solid basis for these developments. Those models are based on continuum mechanics, and consider Plastic Damage Model to simulate concrete behavior. Within this context, this paper presents a new methodology to calculate damage variables evolution; the proposed approach is based in the Lubliner/Lee/Fenves formulation and provides closed-form expressions of the compressive and tensile damage variables in terms of the corresponding strains. This methodology does not require calibration with experimental results and incorporates a strategy to avoid mesh-sensitivity. A particular algorithm, suitable for implementation in Abaqus, is described. Mesh-insensitivity is validated in a simple tension example. Accuracy and reliability are verified by simulating a cyclic experiment on a plain concrete specimen. Two laboratory experiments consisting in pushing until failure two 2-D RC frames are simulated with the proposed approach to investigate its ability to reproduce actual monotonic behavior of RC structures; the obtained results are also compared with the aforementioned simplified models that are commonly employed in earthquake engineering.

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1. Introduction

Under severe seismic excitation, structural behavior of buildings and other constructions is highly complex. It involves, among other issues, soil-structure interaction, large strains and displacements, damage, plasticity, and near-collapse behavior. Moreover, in reinforced concrete structures, there are several coupled degradation and failure modes: cracking, crushing and spalling of concrete, yielding and pull-out of tensioned reinforcement, and yielding and buckling of compressed reinforcement. Therefore, in earthquake engineering, advanced numerical simulations based on continuum mechanics are strongly necessary; conversely, over-simplified models are commonly used, as a result of their moderate computational cost. Furthermore, another circumstance makes the

situation more alarming: given the increasing awareness and concern on the huge worldwide seismic risk, earthquake engineering has experienced in last years substantial advances. New design and analysis strategies have been proposed, leading to relevant developments. These developments rely on extensive testing and numerical simulation; nonetheless, as discussed before, an important number of numerical analyses are mainly conducted by using simplified models. Therefore, there is a strong need of verifying the reliability of the new developments by comparison with analyses performed using more advanced simulation tools. Being aware of this circumstance, the authors have started a long-term research activity aiming to clarify this issue and to provide accurate and reliable models that are based on continuum mechanics. This paper presents early results of this research.

Quasi-brittle materials, as concrete, exhibit nonlinear stress-strain response mainly because of micro-cracking. Cracks are oriented as the stress field and generate the failure modes. In tension, failure is localized in a narrow band; stress-strain behavior is characterized by sudden softening accompanied with reduction in the

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Nomenclature

Roman letters. Lower case

$a_c/a_t/b_c/b_t$	dimensionless coefficients in Eqs. (15) and (16)
b	$\varepsilon_c^{pl}/\varepsilon_c^{ch}$ ratio (Eq. (28))
$c/c_1/c_2$	cohesion/coefficients in Eq. (29)
$d/d_c/d_t$	damage variable/compression damage variable/tension damage variable
$f/f_{bo}/f_{co}/f_{cm}/f_{tm}/f_{co}/f_{to}/f_{ck}$	stress strength/biaxial compressive yield strength/uniaxial compressive yield strength/concrete compressive stress strength/concrete tensile stress strength/limit stress of linear compressive branch/limit stress of linear tensile branch/characteristic value of concrete compressive strength
f_y/f_u	steel yield point/ultimate stress
g_c/g_t	compressive/tensile energies per unit volume dissipated by damage along entire deterioration process
h_c/h_t	weighting factors accounting for stiffness recovery
l_{eq}	mesh size (finite element characteristic length)
p	hydrostatic pressure stress
q	Von Mises-equivalent effective stress
r^*	stress state; for uniaxial stress $r^*(\sigma_{11}) = 1$ for tension and $r^*(\sigma_{11}) = 0$ for compression
s_c/s_t	coefficients accounting for stress state and stiffness recovery effects
w/w_c	crack opening/crack opening at fracture

Roman letters. Upper case

D_b	reinforcement bar diameter
$E/E_0/E_{ci}$	modulus of deformation/undamaged modulus of deformation/tangent modulus of deformation of concrete for zero stress
E_s/E_{sh}	steel modulus of elasticity/slope of hardening branch

F	loading function
$G/G_{ch}/G_f$	flow potential/crushing energy per unit area/fracture energy per unit area
H	Mohr-Coulomb yield surface function
I_1	first invariant of stress tensor
J_2/J_3	second/third invariants of deviatoric stress tensor
K_c	ratio of second stress invariants on tensile and compressive meridians

Greek letters

$\varepsilon/\varepsilon_c/\varepsilon_t/\varepsilon^{el}/\varepsilon^{pl}/\varepsilon_{cm}/\varepsilon_{tm}$	strain/compression strain/tensile strain/elastic strain/plastic strain/strain at compressive strength/strain at tensile strength
$\varepsilon_c^{pl}/\varepsilon_t^{pl}/\varepsilon_c^{el}/\varepsilon_t^{el}/\varepsilon_c^{ch}/\varepsilon_t^{ch}/\varepsilon_{0c}^{el}/\varepsilon_{0t}^{el}$	strains at Fig. 5; subindexes “c”, “t”, “0c” and “0t” and refer to compression, tension, undamaged compression and undamaged tension, respectively; superindexes “pl”, “el”, “ch” and “ck” and refer to plastic, elastic, crushing and cracking, respectively
$\varepsilon_{sh}/\varepsilon_u$	steel strain that corresponds to onset of hardening/ultimate strain
ϵ	eccentricity of the plastic potential surface
ϕ	friction angle
θ	Lode similarity angle
ρ	octahedral radius
$\sigma/\sigma_{11}/\sigma_{t0}/\sigma_{c(1)}/\sigma_{c(2)}/\sigma_{c(3)}$	stress/first principal uniaxial stress/uniaxial tensile stress at failure/concrete compressive stress at first/second/third segment
$\overline{\sigma_c}/\overline{\sigma_t}$	effective compressive/tensile cohesion stress
ξ	distance from origin of stress space to stress plan
ψ	dilatancy angle

unloading stiffness. In compression, failure begins usually in the outside and is more complex, involving volumetric expansion, strain localization, crushing, inclined slipping and spalling; stress-strain behavior involves ductile hardening followed by softening and reduction in the unloading stiffness. In mixed stress states, failure depends usually on the ratio between the principal stresses; in tension-compression, failure is generated by the compression of the material that is between the cracks. Noticeably, in tension the behavior is closer to damage than to plasticity; conversely, in compression the participation of plasticity is higher.

Nonlinear concrete response can be represented using plasticity or damage theory. However, none of these formulations alone is able to describe adequately this phenomenon. Plastic models [21,50] might represent realistically the observed deformation in high confined concrete but do not capture the stiffness degradation observed in experiments [32]. Damage-based models [57,58,20] are based on gradual reduction of the elastic stiffness; they can describe the stiffness degradation in tension and low confined compression, but are not suitable to capture the irreversible deformations observed in experiments and the inelastic volumetric expansion in compression. In addition, fracture propagation can be represented by embedded crack models, where standard FEM interpolations are enriched with strain or displacement discontinuities [12,72,43]. These models can be used for high strain localization problems (fracture).

It is being widely accepted that coupling between damage and plasticity models is essential to capture the nonlinear behavior of concrete [63]. Plasticity for concrete can be described with isotropic hardening; however, damage in many cases is not isotropic but

has preferential directions [30]. Some plastic isotropic damage models have been proposed, e.g. [54,47,16,70,45,42,32,63]; these models have shown good performance in capturing concrete behavior in tests on full-scale structures [30,63]. Anisotropy can be added to the damage to capture the anisotropy feature of concrete both for compression and tension. Although anisotropic damage models are complex and coupling with plasticity in the application to practical engineering is not straightforward, researchers have investigated this issue and proposed plastic anisotropic damage models, among others [67,73,44,35,59,37,17,38,22,82,2]. Even if isotropic damage is a simplified assumption, it is considered in this work because of its simplicity and sufficient accuracy.

Coupled damage and plasticity models for concrete differ mainly in the coupling method and the damage evolution law. In the implicit methods [54,62,70], coupling is embedded in yield and damage criteria; damage evolution law is also implicit. Other researchers describe coupling using a single function. In this context, [47,48] use a yield function; damage measure can be based on some criteria or by postulating damage variables law. This function can be also interpreted as a damage loading [29]; the damage evolution law shall be imposed.

Damage evolution law plays an important role in any damage model, particularly when this law is imposed. A number of researchers have proposed different damage evolutions laws. Most of them are based on splitting damage into compressive and tensile parts and each one is determined separately by its evolution law; total damage is calculated with some combination rules e.g. [29,13,2,52]. Few evolution laws are based on general formulae

for calculating damage in compression and tension [36]. In most of the available damage evaluating formulae the parameters need to be calibrated experimentally [58,29,13,36,2]; other approaches use empirical formulation with some assumptions on stress invariants [83] or with iterative procedures aiming to fit experimental results [52].

After these investigations, this paper proposes a new approach for obtaining damage variables. The need of proposing a new methodology arises from their advantages:

- Is based on the formulation by Lubliner and Lee/Fenves [53,47], which is the base for the Abaqus plastic damage model for concrete. The proposed approach modifies this formulation and obtains closed-form expressions of the damage variables in terms of the corresponding strains; these expressions are derived after integration of concrete fracture and crushing energy.
- No calibration with experimental results is required.
- Implementation is particularly easy.
- Results are insensitive of mesh size, since a strategy aiming to avoid mesh-dependency is incorporated.

All these advantages make this methodology well suited for practical applications. Noticeably, the default Abaqus model based on the Lubliner/Lee/Fenves approach requires that the uniaxial damage variables (i.e. the damage evolution) are provided by the user, and no guidelines are provided. Therefore, defining the damage evolution is a big concern for any user of the Concrete Damage Plastic Model. In the proposed model, constant mesh size is required in the elements corresponding to the same material. If elements with different size are employed for a single material, their parameters should be defined individually for each size following the proposed methodology.

This approach can consider any concrete constitutive law, either empirical (e.g. like formulations commonly recommended by design codes) or directly based on particular experiments; noticeably, any such constitutive law should account for mesh-sensitivity. In this paper, a particular algorithm that uses laws based on European recommendations is derived; in this case, the only input parameters are concrete compressive strength and mesh size. This algorithm is implemented in the software package Abaqus [3]; could be also implemented in other computer codes that contain Plastic Damage Model and that require values of damage variables.

Mesh-insensitivity of the proposed methodology is validated in a simple tension example. Accuracy and reliability are verified by simulating, with the said particular algorithm, a cyclic experiment on plain concrete specimens. Furthermore, two laboratory experiments are described with the proposed approach to investigate its capability to reproduce the behavior of reinforced concrete (RC) framed structures. These experiments are monotonic pushing tests of 2-D RC frames. Obtained numerical results are also compared with those from simplified models that are most commonly utilized in earthquake engineering research and practice. Noticeably, the objective of this comparison is not to highlight the superior quality of the proposed approach, but to point out that using over-simplified models can led to significant errors.

This work considers full bonding between concrete and steel; it is simulated by the “embedded element technique” [49]. This formulation has proven satisfactory for monotonic behavior of reinforced concrete. Conversely, cyclic comportment of RC structures cannot be adequately reproduced with this assumption, because actual bonding behavior involves sliding, thus leading to pinching effects. For that reason, research aiming to incorporate other formulations for describing bonding is currently in progress. Further long-term research will involve extensive use of the derived

models for verification and grounding of commonly used strategies in earthquake engineering and for proposal of new formulations.

2. Concrete Plastic Damage Model

As discussed in the Introduction, the structural behavior of RC structures is highly complex, because the joint operation of concrete and steel. Concrete behavior is brittle, but, under stress reversal, tensile cracks might close, then broken parts being reassembled. Conversely, steel behavior is ductile, with extremely rare fractures, and broken parts cannot be reunited. Therefore, concrete behavior can be better described with damage models, whereas plasticity models better represent steel behavior. Nevertheless, since steel brings additional ductility, the behavior of concrete belonging to reinforced concrete can be even better described with models that combine damage and plasticity. These models are particularly well suited for reproducing failure modes that are based on tensile cracking and compression crushing. In this paper, steel behavior is simulated with a uniaxial plasticity model and concrete is described with a multiaxial model that considers parallel combination of scalar (isotropic) damaged elasticity and non-associated multi-hardening plasticity. This model is termed as “Concrete Plastic Damage Model” (CPDM) along this paper. Was proposed for monotonic, cyclic and dynamic behavior by [53] and was further developed by [47]. This model shows good performance in primarily uniaxial and biaxial stress states, but should not be used in case of significant triaxial compressive stresses.

Fig. 1 displays uniaxial stress-strain plots typical of plasticity, damage and damage-plasticity models; loading branches are represented with solid thick lines and unloading/reloading branches are plotted with dashed thin lines. E_0 is the initial (undamaged) elastic stiffness (deformation modulus), and ϵ^{el} and ϵ^{pl} are the elastic (recoverable) and plastic (irrecoverable) strain, respectively. Fig. 1 shows that damage generates stiffness degradation [66] since the slope of unloading/reloading branch is $(1 - d)E_0$ where d is a damage variable ranging between 0 (no damage) and 1 (destruction).

For uniaxial compression and tension, the stress-strain relation under uniaxial loading in the damage-plasticity behavior displayed in Fig. 1c, can be written as:

$$\sigma_c = (1 - d_c)E_0(\epsilon_c - \epsilon_c^{\text{pl}}) \quad (1)$$

$$\sigma_t = (1 - d_t)E_0(\epsilon_t - \epsilon_t^{\text{pl}}) \quad (2)$$

Subindexes c and t refer to compression and tension, respectively.

For uniaxial cyclic loading-unloading conditions, the damage plasticity model assumes that the degradation in the elastic stiffness is given by

$$E = (1 - d)E_0 \quad (3)$$

In Eq. (3), E is the reduced tangent stiffness and d is a scalar degradation variable, which is a function of stress state and of compression and tension damage variables (d_c and d_t , respectively):

$$1 - d = (1 - s_t d_c)(1 - s_c d_t) \quad (4)$$

In Eq. (4), s_c and s_t are dimensionless coefficients accounting for stress state and stiffness recovery effects, being given by

$$s_c = 1 - h_c(1 - r^*(\sigma_{11})) \quad (5)$$

$$s_t = 1 - h_t r^*(\sigma_{11}) \quad (6)$$

In Eqs. (5) and (6), σ_{11} is the first principal uniaxial stress (positive for tension), r^* is a stress state parameter being $r^*(\sigma_{11}) = 1$ for tension and $r^*(\sigma_{11}) = 0$ for compression, and h_c and h_t are

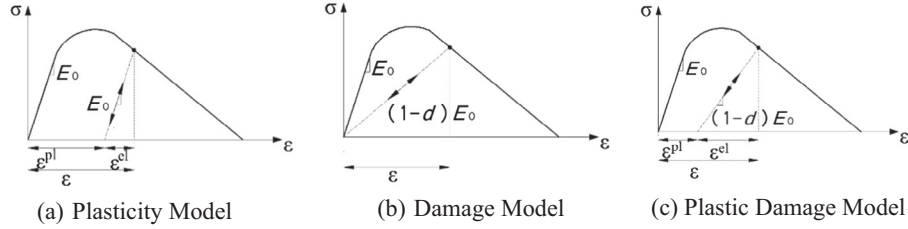


Fig. 1. Representation of CPDM.

weighting factors ranging between 0 and 1. Factor h_c accounts for re-closing of cracks after tension-compression reversal; h_t represents recovery of crushed concrete after compression-tension reversal. In this work, $h_c = 0.9$ and $h_t = 0$ is assumed; this means that 90% of the cracks close upon tension-compression reversal and the crushed concrete does not experience any recovery. Eqs. (5) and (6) show that s_c and s_t also range between 0 and 1.

For a better understanding of the effect of s_c and s_t coefficients, Fig. 2 displays plots of uniaxial stress-strain loading-unloading behavior. The initial elastic branch with slope E_0 reaches the descending branch (Fig. 5.b) at peak point 1, then cracking begins; later, unloading starts at point 2. At that point, there is no compression damage and $d_c = 0$, $r^* = 1$, and $s_c = 1$; therefore, Eq. (4) shows that $d = d_t$. Consequently, the linear unloading branch has slope $(1 - d_t)E_0$. In the way to stress reversing point 3, cracks begin closing. After point 3, $r^* = 0$, $s_c = 1 - h_c$, $s_t = 1$, $d_c = 0$, and Eq. (4) shows that $d = (1 - h_c)d_t$. Therefore, the slope of the ongoing compression segment of the branch depends on parameter h_c ; three options are plotted in Fig. 2: (i) $h_c = 0$ (no crack is closed) with slope $(1 - d_t)E_0$, (ii) $h_c = 0.5$ (half of the cracks are closed) with slope $(1 - 0.5 d_t)E_0$, (iii) $h_c = 1$ (all cracks are closed) with slope E_0 . Noticeably, in the third option ($h_c = 1$), there is no compressive strength reduction. At point 4, an unloading branch arises; at that point, $r^* = 0$, $s_c = 1 - h_c$, $s_t = 1$, and Eq. (4) shows that $1 - d = (1 - d_c)[1 - (1 - h_c)d_t] = 1 - d_c$. Point 5 correspond again to stress reversal; after it, provided that $h_t = 0$, the slope of the ongoing branch is equal to $(1 - d_t)(1 - d_c)E_0$. Point 6 is the peak for the reduced tensile strength; after it, cracking reinitiates and a new descending branch is generated.

For multiaxial condition, the stress-strain relationship is given by:

$$\sigma = (1 - d)D_0^{\text{el}} : (\epsilon - \epsilon^{\text{pl}}) \quad (7)$$

In Eq. (7), D_0^{el} is the elastic stiffness tensor, and σ and ϵ are the stress and strain tensors, respectively. Scalar damage variable d

keeps same meaning than for uniaxial condition, although replacing scalar factor r^* with a multiaxial one [47].

Regarding plasticity model, yield condition is based on the loading function F proposed in [53] with modifications suggested by [47] to account for different tension and compression strength evolution.

$$F = \frac{1}{1 - \alpha}(q - 3\alpha p + \beta \langle \sigma_{\max} \rangle - \gamma - \langle \sigma_{\max} \rangle) - \overline{\sigma}_c = 0 \quad (8)$$

$$\alpha = \frac{(f_{b0}/f_{c0}) - 1}{2(f_{b0}/f_{c0}) - 1}; \quad \beta = \frac{\overline{\sigma}_c}{\overline{\sigma}_t}(1 - \alpha) - (1 + \alpha); \quad \gamma = \frac{3(1 - K_c)}{2K_c - 1} \quad (9)$$

In Eqs. (8) and (9), $\langle \cdot \rangle$ is the Macaulay bracket, p is the hydrostatic pressure stress, q is the Von Mises-equivalent effective stress (effective stress accounts for stress divided by $1 - d$), and f_{b0} and f_{c0} are the biaxial and uniaxial compressive yield strengths, respectively; since $f_{b0} \geq f_{c0}$, α ranges between 0 ($f_{b0} = f_{c0}$) and 0.5 ($f_{b0} \gg f_{c0}$). $\sigma_{\max}$ is the maximum principal effective stress, and $\overline{\sigma}_c$ and $\overline{\sigma}_t$ are the effective compressive and tensile cohesion stress, respectively. $\overline{\sigma}_c$ and $\overline{\sigma}_t$ are defined as $\overline{\sigma}_c = \sigma_c/(1 - d_c)$ and $\overline{\sigma}_t = \sigma_t/(1 - d_t)$. K_c is the ratio of second stress invariants on tensile and compressive meridians.

The plasticity model assumes non-associated potential plastic flow. The flow potential G is the Drucker-Prager hyperbolic function given by:

$$G = \sqrt{(\epsilon \sigma_{t0} \tan \psi)^2 + q^2} - p \tan \psi \quad (10)$$

In Eq. (10), σ_{t0} is the uniaxial tensile stress at failure, ϵ is the eccentricity of plastic potential surface, and ψ is the dilatancy angle measured in $p - q$ (deviatory) plan at high confining pressure.

As discussed previously, K_c is the ratio between the magnitudes of deviatoric stress in uniaxial tension and compression; K_c ranges between 0.5 (Rankine yield surface) and 1 (Von Mises). In this

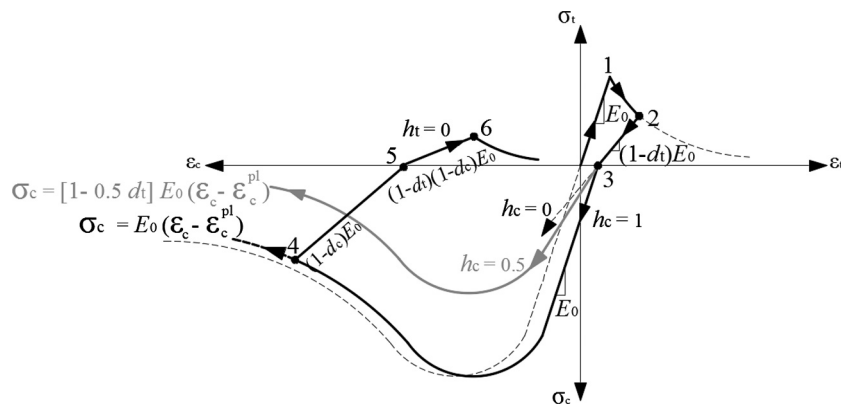


Fig. 2. Uniaxial loading-unloading law.

study, K_c is obtained from the Mohr-Coulomb yield surface function in cylindrical coordinates [66]:

$$H(\rho, \xi, \theta, \phi, c) = \sqrt{2}\xi \sin\phi + \sqrt{3}\rho \cos\theta - \rho \sin\theta \sin\phi - \sqrt{6}c \cos\phi = 0 \quad (11)$$

In Eq. (11), ρ is the octahedral radius, ξ is the distance from the origin of stress space to the stress plan, θ is the Lode similarity angle, ϕ is the friction angle, and c is the cohesion. In Eq. (11), $\rho = \sqrt{2J_2}$, $\xi = I_1/\sqrt{3}$, and $\sin\theta = \frac{3\sqrt{3}J_3}{2J_2^{3/2}}$; I_1 is the first invariant of stress tensor and J_2 and J_3 are the second and third invariants of deviatoric stress tensor, respectively.

For $\xi = 0$ and $\theta = \pm\pi/6$ (negative/positive for tension/compression meridian plans), the magnitudes of deviatoric stress in uniaxial compression and tension at yield (ρ_{c0} and ρ_{t0}) and K_c are

$$\rho_{c0} = \frac{2c\sqrt{6}\cos\phi}{3 - \sin\phi} \quad \rho_{t0} = \frac{2c\sqrt{6}\cos\phi}{3 + \sin\phi} \quad K_c = \frac{\rho_{t0}}{\rho_{c0}} = \frac{3 - \sin\phi}{3 + \sin\phi} \quad (12)$$

By assuming that $\phi = 32^\circ$ [66], last equation in (12) shows that $K_c = 0.7$. Fig. 3 represents the yield surface in deviatoric plan for several values of K_c ranging from 0.5 to 1. CM and TM account for compression/tension meridian plans, respectively.

Eqs. (10), (8) and (9) show that the concrete behavior depends on four constitutive parameters K_c , ψ , f_{b0}/f_{c0} , ϵ ; it can be assumed that $\psi = 13^\circ$ [80]. Table 1 describes the values used in this study.

3. Proposed methodology for calculating damage variables

3.1. General description

The proposed approach for calculating the damage variables starts from definition of compressive and tensile variables as the portion of normalized energy dissipated by damage:

$$d_c = \frac{1}{g_c} \int_0^{\epsilon_c^{ch}} \sigma_c d\epsilon_c^{ch} \quad d_t = \frac{1}{g_t} \int_0^{\epsilon_t^{ck}} \sigma_t d\epsilon_t^{ck} \quad (13)$$

In Eq. (13), ϵ_c^{ch} and ϵ_t^{ck} are the crushing and cracking strains respectively, see Fig. 5. Normalization coefficients g_c and g_t represent the energies per unit volume dissipated by damage along the entire deterioration process:

$$g_c = \int_0^\infty \sigma_c d\epsilon_c^{ch} \quad g_t = \int_0^\infty \sigma_t d\epsilon_t^{ck} \quad (14)$$

Eqs. (13) and (14) show that d_c and d_t range between 0 (no damage) and 1 (destruction). Fig. 4 describes the meaning of g_c and g_t .

Noticeably, the energies per unit area and per unit volume are related by $g_c = G_{ch}/l_{eq}$ and $g_t = G_F/l_{eq}$; G_{ch} and G_F are material

Table 1
Parameters of CPDM.

K_c	Ψ (°)	f_{b0}/f_{c0}	ϵ
0.7	13	1.16	0.1

parameters defined as crushing and fracture energies and l_{eq} is the characteristic length of the element (Section 3.2).

Relation between compressive and tensile stress and, respectively, crushing and cracking strain, is established, according [53], as:

$$\sigma_c = f_{c0} [(1 + a_c) \exp(-b_c \epsilon_c^{ch}) - a_c \exp(-2b_c \epsilon_c^{ch})] \quad (15)$$

$$\sigma_t = f_{t0} [(1 + a_t) \exp(-b_t \epsilon_t^{ck}) - a_t \exp(-2b_t \epsilon_t^{ck})] \quad (16)$$

In Eqs. (15) and (16), f_{c0} and f_{t0} are the compressive and tensile stresses that correspond to zero crushing ($\epsilon_c^{ch} = 0$) and to onset of cracking ($\epsilon_t^{ck} = 0$), respectively; see Fig. 4. As well, a_c , a_t , b_c and b_t are dimensionless coefficients to be determined. Replacing Eqs. (15) and (16) in Eq. (14), provides the following relations among g_c and g_t and such coefficients:

$$g_c = \frac{f_{c0}}{b_c} \left(1 + \frac{a_c}{2}\right) \quad g_t = \frac{f_{t0}}{b_t} \left(1 + \frac{a_t}{2}\right) \quad (17)$$

Coefficients b_c and b_t can be obtained by replacing $g_c = G_{ch}/l_{eq}$ and $g_t = G_F/l_{eq}$ in Eq. (17):

$$b_c = \frac{f_{c0} l_{eq}}{G_{ch}} \left(1 + \frac{a_c}{2}\right) \quad b_t = \frac{f_{t0} l_{eq}}{G_F} \left(1 + \frac{a_t}{2}\right) \quad (18)$$

By substituting results (15)–(17) in Eq. (13), the proposed tensile and compressive damage functions are derived:

$$d_c = 1 - \frac{1}{2 + a_c} [2(1 + a_c) \exp(-b_c \epsilon_c^{ch}) - a_c \exp(-2b_c \epsilon_c^{ch})] \quad (19)$$

$$d_t = 1 - \frac{1}{2 + a_t} [2(1 + a_t) \exp(-b_t \epsilon_t^{ck}) - a_t \exp(-2b_t \epsilon_t^{ck})] \quad (20)$$

Therefore, provided that coefficients a_c and a_t are not zero:

$$\exp(-b_c \epsilon_c^{ch}) = \frac{1}{a_c} \left[1 + a_c - \sqrt{1 + a_c(2 + a_c)d_c}\right] \quad (21)$$

$$\exp(-b_t \epsilon_t^{ck}) = \frac{1}{a_t} \left[1 + a_t - \sqrt{1 + a_t(2 + a_t)d_t}\right] \quad (22)$$

By zeroing derivatives of σ_c and σ_t (Eqs. (15) and (16), respectively) with respect to, respectively, crushing and cracking strain, maximum values f_{cm} and f_{tm} (Fig. 5) are obtained:

$$f_{cm} = \frac{f_{c0}(1 + a_c)^2}{4a_c} \quad f_{tm} = \frac{f_{t0}(1 + a_t)^2}{4a_t} \quad (23)$$

Eq. (23) provide:

$$a_c = 2(f_{cm}/f_{c0}) - 1 + 2\sqrt{(f_{cm}/f_{c0})^2 - (f_{cm}/f_{c0})} \quad (24)$$

$$a_t = 2(f_{tm}/f_{t0}) - 1 + 2\sqrt{(f_{tm}/f_{t0})^2 - (f_{tm}/f_{t0})} \quad (25)$$

These developments complete the description of the proposed methodology. Parameters a_c , a_t , b_c and b_t can be determined from Eqs. (24), (25) and (18) in terms of f_{cm} , f_{c0} , f_{tm} , f_{t0} , l_{eq} , G_{ch} and G_F . Then, the damage functions can be calculated according Eqs. (19) and (20). An implementation of the proposed approach using the concrete behavior described in next subsection is presented in Section 3.3.

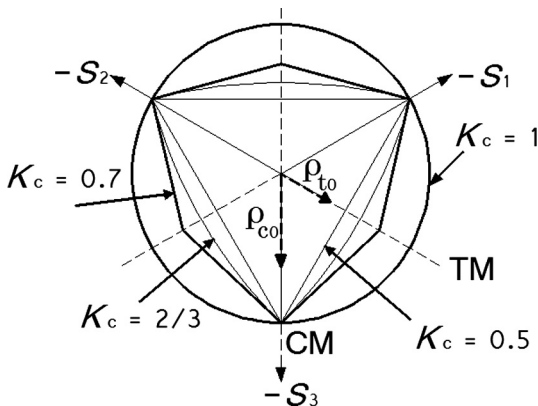


Fig. 3. Yield surface in the deviatoric plan for several values of K_c .

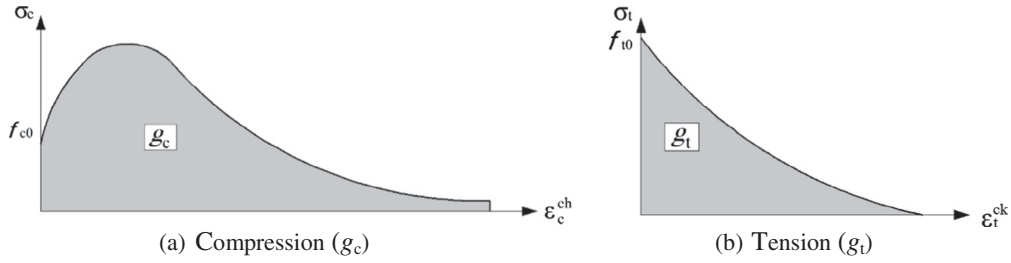


Fig. 4. Parts of energy dissipated by damage.

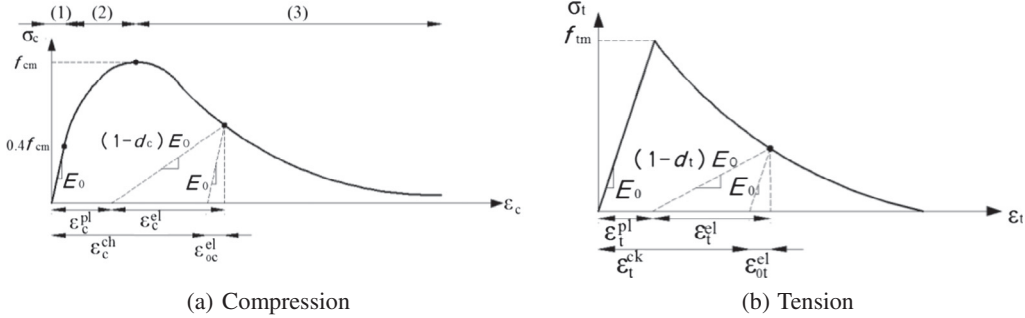


Fig. 5. Assumed uniaxial model of concrete behavior.

3.2. Uniaxial concrete behavior

This subsection describes the concrete uniaxial stress-strain law that is selected for implementation. Fig. 5a and b explain the compressive and tensile models, respectively; noticeably, tensile model is smeared, i.e. strain includes both crack opening and actual tensile strain between cracks. Fig. 5 displays the constitutive laws (thick solid lines) and the unloading/reloading branches (thin dashed lines). Fig. 5 corresponds to the damage-plasticity behavior depicted in Fig. 1c. The ascending compressive segments in Fig. 5a follow the Model Code recommendations [19] and the descending segment is engendered as [45]. Tensile stress-strain relation consists of an initial linear segment and a nonlinear descending branch [81,76], as shown in Fig. 5b. Both compressive and tensile descending branches are generated to ensure nearly mesh-independency; the regularization approach is based on selecting the softening branches of concrete constitutive laws depending on mesh size. In Fig. 5, f_{cm} and f_{tm} represent compressive and tensile stress strength, respectively; corresponding strains are ϵ_{cm} and ϵ_{tm} , respectively. According to [19], it is assumed that $\epsilon_{cm} = 0.0022$, $f_{cm} = f_{ck} + 8$ (f_{ck} is the characteristic value of concrete compressive strength), and $f_{tm} = 0.3016f_{ck}^{2/3}$; these stresses and the deformation modulus are expressed in MPa. In Fig. 5a, ϵ_c^{ch} and ϵ_{0c}^{el} are the crushing and elastic undamaged components of strain; ϵ_c^{pl} and ϵ_c^{el} are the plastic and elastic damaged components. In Fig. 5b, ϵ_t^{ck} and ϵ_{0t}^{el} are the cracking and elastic undamaged strain components; ϵ_t^{pl} and ϵ_t^{el} are the plastic and elastic damaged components.

First segment in Fig. 5a is linear, $\sigma_{c(1)} = E_0 \epsilon_c$, reaching $0.4 f_{cm}$; second (ascending) segment (in between $0.4 f_{cm}$ and f_{cm}) is quadratic [19]:

$$\sigma_{c(2)} = \frac{E_{ci} \frac{\epsilon_c}{f_{cm}} - \left(\frac{\epsilon_c}{\epsilon_{cm}} \right)^2}{1 + \left(E_{ci} \frac{\epsilon_{cm}}{f_{cm}} - 2 \right) \frac{\epsilon_c}{\epsilon_{cm}}} f_{cm} \quad (26)$$

In Eq. (26), E_{ci} is the modulus of deformation of concrete for zero stress, given by $E_{ci} = 10000f_{cm}^{1/3}$ and $E_0 = (0.8 + 0.2 f_{cm}/88) E_{ci}$

(in MPa) [19]. In the initial linear branch, E_0 is the secant modulus that corresponds to $0.4 f_{cm}$ stress.

Third (descending) segment is given by:

$$\sigma_{c(3)} = \left(\frac{2 + \gamma_c f_{cm} \epsilon_{cm}}{2 f_{cm}} - \gamma_c \epsilon_c + \frac{\epsilon_c^2 \gamma_c}{2 \epsilon_{cm}} \right)^{-1} \quad (27)$$

$$\gamma_c = \frac{\pi^2 f_{cm} \epsilon_{cm}}{2 \left[l_{eq} - 0.5 f_{cm} \left(\epsilon_{cm} (1 - b) + b \frac{f_{cm}}{E_0} \right) \right]^2} \quad b = \frac{\epsilon_c^{pl}}{\epsilon_c^{ch}} \quad (28)$$

In Eqs. (27) and (28), G_{ch} is the crushing energy per unit area [45], and l_{eq} is the characteristic length, which depends on the mesh size, the type of finite element and the crack direction [64,45]. Assuming idealized behavior of single band of cracks, the characteristic length can be determined after the mesh size; [60] relate the crack width with the square root of the finite element area for 2D elements. In this work, brick solid elements are utilized; the characteristic length is taken as the volume divided by the largest face area.

Based on experimental observations, $b = 0.9$ (Eq. (28)) can be initially assumed. After calculating the damage variables, the average value of b along the relevant strain range is obtained; iterative calculations are performed until reaching convergence. The final value of b affects the softening branch of the compressive stress-strain relation (Eqs. (27) and (28)); therefore, the dissipated crushing energy will be changed. Anyway, this effect is not very intense.

Eq. (27) shows that the descending branch approaches asymptotically zero; therefore, a fictitious maximum strain shall be selected for calculation purposes. The maximum strain value shall fulfill that the crushing energy in Eq. (32) is equal to the area under the corresponding compressive stress-strain law multiplied by the characteristic length.

Regarding tensile behavior, the ratio between tensile stress $\sigma_t(w)$ (for crack width w) and maximum tensile strength f_{tm} , is given by [40]:

$$\frac{\sigma_t(w)}{f_{tm}} = \left[1 + \left(c_1 \frac{w}{w_c} \right)^3 \right] e^{-c_2 \frac{w}{w_c}} - \frac{w}{w_c} (1 + c_1^3) e^{-c_2} \quad (29)$$

In Eq. (29), $c_1 = 3$, $c_2 = 6.93$ [40], and w_c is the critical crack opening. Eq. (29) shows that $\sigma_t(0) = f_{tm}$ and $\sigma_t(w_c) = 0$. Therefore, w_c can be considered as the fracture crack opening. Eq. (30) [40] relates w_c with the tensile strength and fracture energy G_F per unit area:

$$w_c = 5.14 G_F / f_{tm} \quad (30)$$

According to [19], G_F (N/mm) can be calculated as

$$G_F = 0.073 f_{cm}^{0.18} \quad (31)$$

In Eq. (31), f_{cm} is expressed in MPa. The ratio between crushing and fracture energies can be assumed proportional to square of the ratio between compressive and tensile strengths [65]:

$$G_{ch} = \left(\frac{f_{cm}}{f_{tm}} \right)^2 G_F \quad (32)$$

In this work, the actual crack spacing is not studied, it has been assumed that there is a single crack per element. This supposition is suitable for global-purpose simulation. After this assumption, in the descending segment of the tensile stress-strain curve (Fig. 5b), the strain can be obtained in terms of the crack opening from the following kinematic relation:

$$\varepsilon_t = \varepsilon_{tm} + w/l_{eq} \quad (33)$$

This subsection describes the concrete uniaxial concrete behavior; next subsection describes their implementation in the proposed approach.

3.3. Implementation

Following the formulation described in the previous section, coefficients a_c , a_t , b_c and b_t (Section 3.1) can be determined. Coefficient a_c is obtained by replacing $f_{c0} = 0.4 f_{cm}$ (Fig. 5a) in Eq. (24). Fig. 5b shows that $f_{tm} = f_{t0}$, then Eq. (25) shows that $a_t = 1$. Coefficient b_c can be determined from Eq. (18) by replacing $f_{c0} = 0.4 f_{cm} = 0.4 (f_{ck} + 8)$ MPa. Coefficient b_t is determined by replacing $a_t = 1$ and $f_{t0} = f_{tm} = 0.3016 f_{ck}^{2/3}$ [19] in Eq. (18). Therefore:

$$a_c = 7.873 \quad a_t = 1 \quad b_c = \frac{1.97(f_{ck} + 8)}{G_{ch}} l_{eq} \quad b_t = \frac{0.453 f_{ck}^{2/3}}{G_F} l_{eq} \quad (34)$$

In the last two expressions in Eq. (34), f_{ck} is expressed in MPa. After parameters a_c , a_t , b_c and b_t , damage variables d_c and d_t are determined by Eqs. (19) and (20) in terms of crushing and cracking strain, respectively.

A particular implementation of the proposed methodology by using the previously described concrete constitutive law (Section 3.2) is described next. All stress values are in MPa.

1. The input data are the concrete compressive strength f_{ck} , the parameters in Table 1, the mesh size l_{eq} , and the ratio b (eqn. (28)). Initial assumption is $b = 0.9$.
2. Calculate the compressive/tensile stress strength $f_{cm} = f_{ck} + 8/f_{tm} = 0.3016 f_{ck}^{2/3}$.
3. State the strain at compressive stress strength as $\varepsilon_{cm} = 0.0022$.
4. Calculate the initial tangent modulus of deformation of concrete $E_{ci} = 10000 f_{cm}^{1/3}$ and the undamaged modulus of deformation $E_0 = E_{ci} (0.8 + 0.2 \frac{f_{cm}}{88})$.
5. Calculate the fracture/crushing energy (N/mm) $G_F = 0.073 f_{cm}^{0.18}$ / $G_{ch} = (\frac{f_{cm}}{f_{tm}})^2 G_F$.
6. Calculate the critical crack opening $w_c = 5.14 G_F / f_{tm}$.

7. Build the first/second/third segments of the concrete uniaxial compressive law: $\sigma_{c(1)} = E_0 \varepsilon_c / \text{Eq. (26)/Eq. (27)}$. In Eq. (27), strain is bounded; the selected upper bound should fulfill the condition that the crushing energy G_{ch} (Eq. (32)) is reached.
8. Build the first/second segment of the concrete uniaxial tensile law: $\sigma_{t(1)} = E_0 \varepsilon_t / \text{Eqs. (29) and (33)}$.
9. Calculate the damage parameters according Eq. (34): $a_c = 7.873$; $a_t = 1$; $b_c = \frac{1.97(f_{ck} + 8)}{G_{ch}} l_{eq}$; $b_t = \frac{0.453 f_{ck}^{2/3}}{G_F} l_{eq}$.
10. Calculate the compressive/tensile damage variables (damage evolution) according to Eqs. (19)/(20).
11. Calculate the compressive and tensile plastic strains as indicated in Fig. 5: $\varepsilon_c^{pl} = \varepsilon_c^{ch} - \sigma_c d_c / (1 - d_c) E_0$, $\varepsilon_t^{pl} = \varepsilon_t^{ck} - \sigma_t d_t / (1 - d_t) E_0$.
12. Calculate the average value of ratio $b = \frac{\varepsilon_c^{pl}}{\varepsilon_c^{ch}}$ and compare with the assumption in step 1. Repeat until reaching convergence.

Once convergence is reached, the major output are curves of compressive/tensile stress and damage variables vs. crushing/cracking strain, respectively; in Section 3.4, Fig. 7 presents some examples. These plots constitute the input of the implementation in any model describing the global structural behavior. This algorithm is suited for software package Abaqus [3,51,18]. Noticeably, the proposed approach can be also fitted for other computer codes and for other concrete formulations, either empirical (code-type) or directly based on experiments.

Time integration follows an implicit formulation, and the global algorithm is generated by imposing energy balance. In this work, analyses are conducted for large displacements, although not for large strains.

3.4. Mesh-insensitivity verification

When the material exhibits softening, finite element size influences significantly the entire model behavior due to localization since the dissipated energy decreases upon mesh refinement. This can be solved by the so-called mesh regularization techniques. One of the simplest remedy is the Crack Band Method; it uses energy-based scaling of the softening part of the stress-strain relation [9]. This technique can be considered as a simple and effective method for practical engineering analysis and has been implemented in many concrete plastic damage models, among others [32,22]. More advanced techniques are the so-called Non-Local approaches; they are based on introducing non-locality in the constitutive model. This non-locality can be incorporated into an integral format [10,11]; this approach has been implemented in different plastic damage models [33,63]. Another strategy for incorporating non-locality is including higher-order deformation gradients in the model [69]. This has been implemented in several plastic damage models [5,1].

As the model of [53] is the base of this work and this model uses the fracture energy-based regularization to describe the softening in tension and compression, the crack band technique is considered. Then both fracture and crushing energies are scaled in relation to the finite element size. The main assumption in this technique is that damage is localized in a single row of elements; this is true for the physical failure mechanism of concrete in tension but is not exactly right for compression. Uniaxial compression tests show a great relation between boundary conditions, concrete strength, size, and failure mechanism [46,77]. It varies between cone failure and vertical splitting modes. Cone failure occurs when friction at top and bottom sections restrains lateral expansion; this mode is characterized by inclined slip bands at each corner

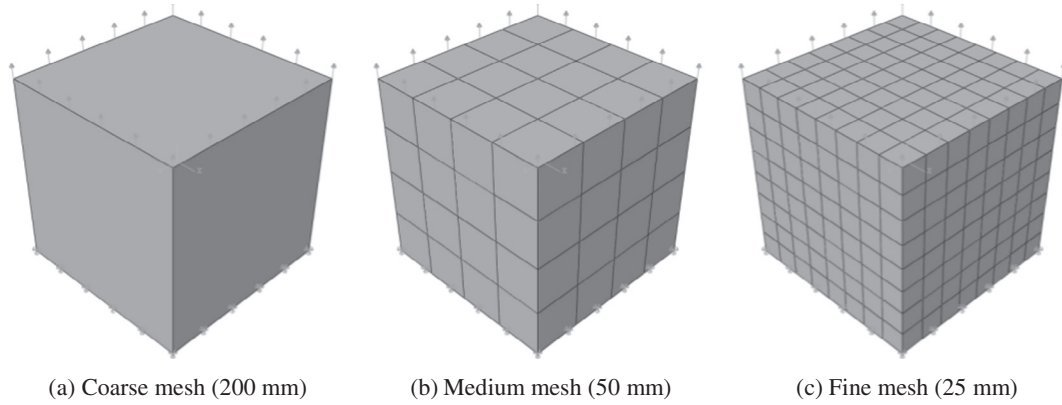


Fig. 6. Uniaxial tension example.

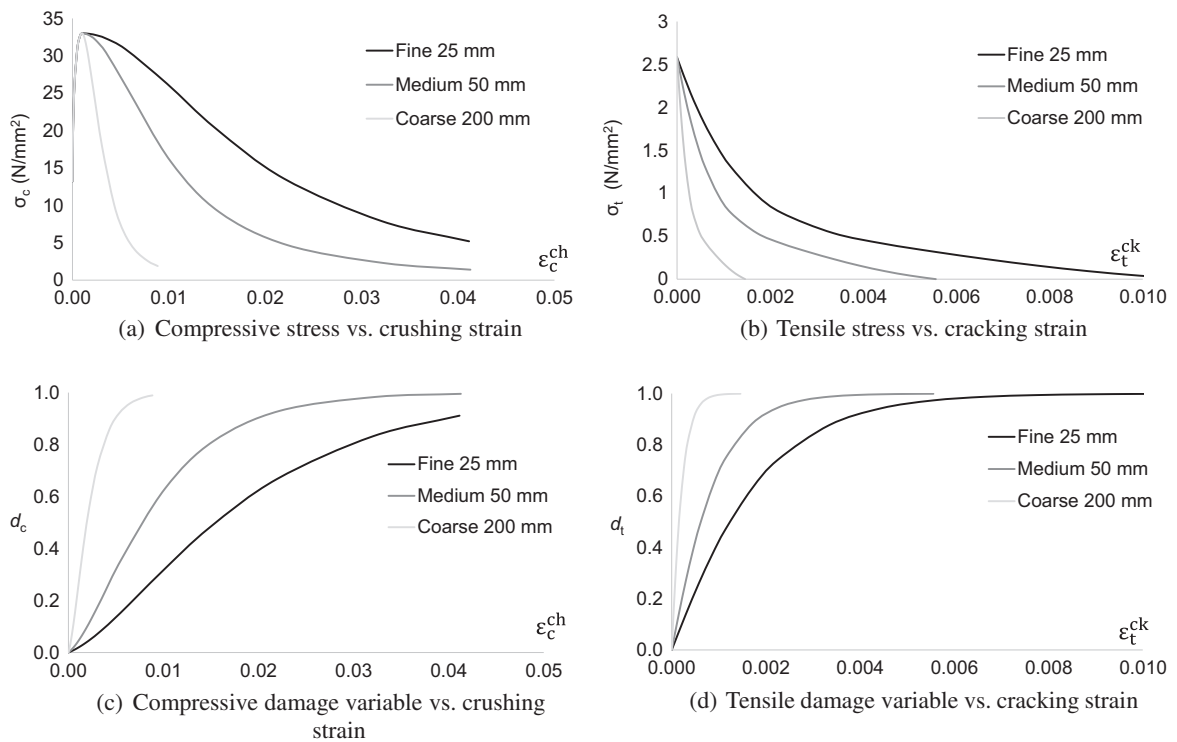


Fig. 7. Results of the proposed algorithm for the uniaxial tension example using coarse, medium and fine meshes.

and crushing in the middle zone for very stocky specimens. Vertical splitting mode can happen for highly slender specimens when the effect of friction is negligible. Intermediate failure scenarios can occur for different degrees of friction and slenderness; they are characterized mainly by inclined shear band zone. Regarding damage localization under compression, experimental observation by [46,76,41] showed strain localization; analytical investigation by [56] showed that the concept of fracture energy in tension holds true for compression. This gives the evidence to use the fracture energy-based regularization in describing the softening of concrete under compression [78].

In this subsection the capacity of proposed methodology to overcome mesh insensitivity is verified for a uniaxial tension problem. Given the above considerations, it is apparent that the uniaxial compression case cannot be verified in the same way. For practical engineering applications, behavior is mainly controlled by bending and there will be mixed stresses states with high stress gradients; therefore, the possible inaccuracies will have little effect on overall results.

This subsection describes the application of the proposed methodology to the problem of uniaxial tension of a $200 \text{ mm} \times 200 \text{ mm} \times 200 \text{ mm}$ cube, as described in Fig. 6. The cube is discretized with a uniform mesh of 3D 8-node hexahedron solid finite elements (C3D8R). Fig. 6 displays three sketches of the cube discretized with a coarse mesh ($l_{eq} = 200 \text{ mm}$, one element), a medium mesh ($l_{eq} = 50 \text{ mm}$, 64 elements) and a fine mesh ($l_{eq} = 25 \text{ mm}$, 512 elements), respectively. The objective of this analysis is to verify the allegedly low sensitivity to mesh size.

This problem is analyzed with the algorithm described in Section 3.2. Table 2 displays the values of the parameters for each mesh size. Any of the values of parameter b in Table 2 is the result of an independent iterative process starting from 0.9. For consistency, the final value is independent on this initial assumption.

The algorithm is implemented in Abaqus code [3]. The displacement is applied incrementally; for coarse/medium/fine meshes, the maximum number of iterations is 4, 6 and 5, respectively.

Fig. 7 presents the plots that constitute the major inputs for this implementation. Fig. 7a and b display plots of compressive/tensile

stress vs. crushing/cracking strain, respectively; Fig. 7c and d display plots of compressive/tensile damage variable vs. crushing/cracking strain, respectively. Fig. 7 shows that the inputs are strongly dependent on the mesh size; conversely, outputs are expected to be almost independent on it. Noticeably, compressive plots are displayed only for information.

Given that the problem under consideration is extremely simple, a closed-form solution can be provided. This solution is derived from the uniaxial tensile constitutive law given by Eq. (29) (Fig. 5b), with the critical crack opening w_c obtained from Eq. (30) ($w_c = 0.273$ mm).

After the implementation in Abaqus of the plots displayed in Fig. 7, Fig. 8 presents the force–displacement plots that are obtained for each mesh size. For comparison purposes, Fig. 8 contains also the aforementioned closed-form solution. Fig. 8 shows a satisfactory agreement among the plots for the three mesh sizes; this observation corroborates the mesh-insensitiveness of the proposed methodology. As well, numerical results match analytical ones, thus confirming the accuracy. Noticeably, the best match is provided by the coarse mesh; this circumstance can be explained because a single element is the closest representation of the closed-form solution. On the other hand, the result of the fine mesh is better than the one of the medium mesh; certainly, further refinements would lead to higher accuracy.

The area under the plots in Fig. 8 (between the peak and the displacement for zero stress in the closed-form solution) is equal to 5.5149, 5.6358 and 5.3965 Nm for coarse, medium and fine meshes, respectively. These quantities represent the fracture work; it can be converted into fracture energy per unit area (G_F) dividing by the area perpendicular to the applied displacement following the regularization method [9]; results are: 0.1379, 0.1409 and 0.1349 N/mm for coarse, medium and fine meshes, respectively. Comparison with the values indicated in Table 2 shows a satisfactory agreement.

3.5. Experimental validation on a specimen test

Accuracy and reliability of the proposed methodology are verified by simulating one experiment with the particular algorithm described in Section 3.3 and with the formulation to determine the damage variables described in [13] (Section 1). The experiment [74] consisted in imposing a cyclic force law to a number of plain concrete cylinders. Fig. 9 displays the imposed force law (in terms of stress).

This experiment had been previously simulated [2,6]. In the work [2] a coupled plasticity–damage model was used; the numerical algorithm was coded and then implemented in the commercial package Abaqus. In the reference [6], this experiment was described with a hysteretic stress–strain model developed for unconfined concrete.

The simulated test refers to a cube element with 125 mm side; it is discretized with a single element. The algorithm is implemented in Abaqus code [3]. The maximum number of iterations is 2, and loading increment is 4×10^{-4} mm. Table 3 displays the values of the parameters.

Fig. 10 displays major outputs of the simulations, compared with the corresponding experimental results. Fig. 10a exhibits

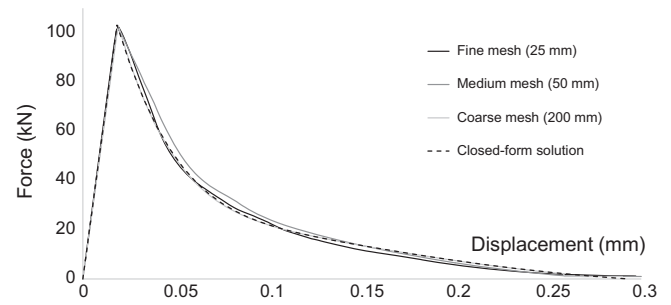


Fig. 8. Force-deflection plots for the uniaxial tension example using coarse, medium and fine meshes.

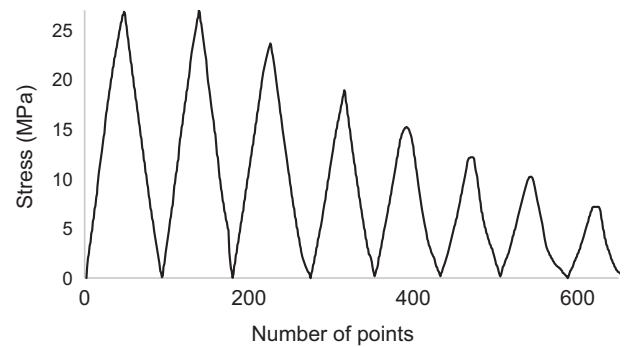


Fig. 9. Input force law for the plain concrete test.

stress–strain plots and Fig. 10b shows the evolution of damage variable (d) in terms of strain. In points “▲” in Fig. 10b, the abscissa corresponds to measured values of strain and the ordinate corresponds to damage determined applying Eq. (3) to the experimental plots in Fig. 10a, as average for each pair of unloading and reloading branches. In the numerical plots in Fig. 10b, the horizontal segments belong to constant damage along unloading and reloading branches.

Fig. 10a shows a satisfactory agreement between numerical and experimental results, particularly in terms of envelope curve; noticeably, envelope results from both models are almost identical (because the same uniaxial constitutive laws have been considered). Fig. 10b shows an adequate agreement between the experimental damage (at the measured strains) and the numerical damage calculated with the proposed approach, given that points “▲” are highly close to the numerical curve.

4. Simulation of monotonic testing of RC frames

4.1. First frame test

This experiment [79] is a quasi-static test consisting of pushing monotonically until failure a laboratory, single-span, two-story, planar RC frame. Fig. 11b and c display front and side views of the tested frame, respectively. Fig. 11a and d exhibit cross sections of columns and beams, respectively. In Fig. 11 dimensions are in mm. Fig. 11b shows that both columns were loaded with constant

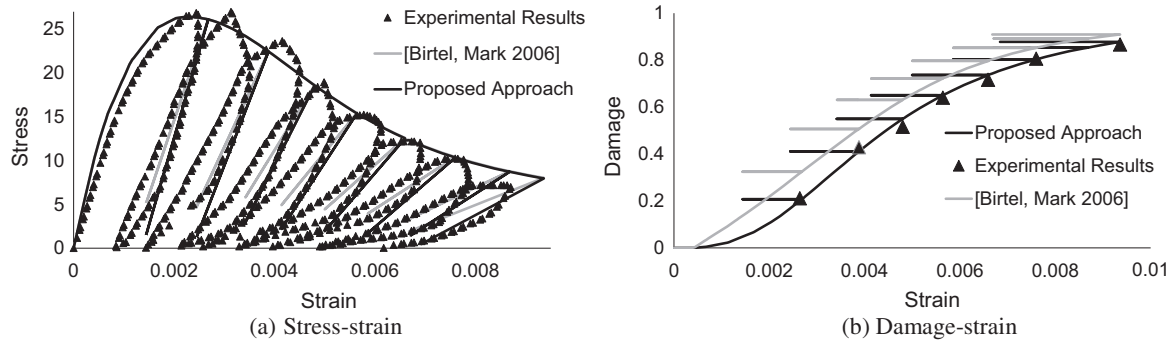
Table 2
Parameters for the mesh-insensitivity verification example.

Mesh	l_{eq} (mm)	f_{ck} (MPa)	f_{cm} (MPa)	f_{tm} (MPa)	G_{ch} (N/mm)	G_F (N/mm)	b	a_c	a_t	b_c	b_t
Coarse	200	25	33	2.58	22.43	0.137	0.6	7.873	1	581	5648
Medium	50	25	33	2.58	22.43	0.137	0.914	7.873	1	145.2	1412
Fine	25	25	33	2.58	22.43	0.137	0.967	7.873	1	72.6	706

Table 3

Parameters for the simulation of the plain concrete test.

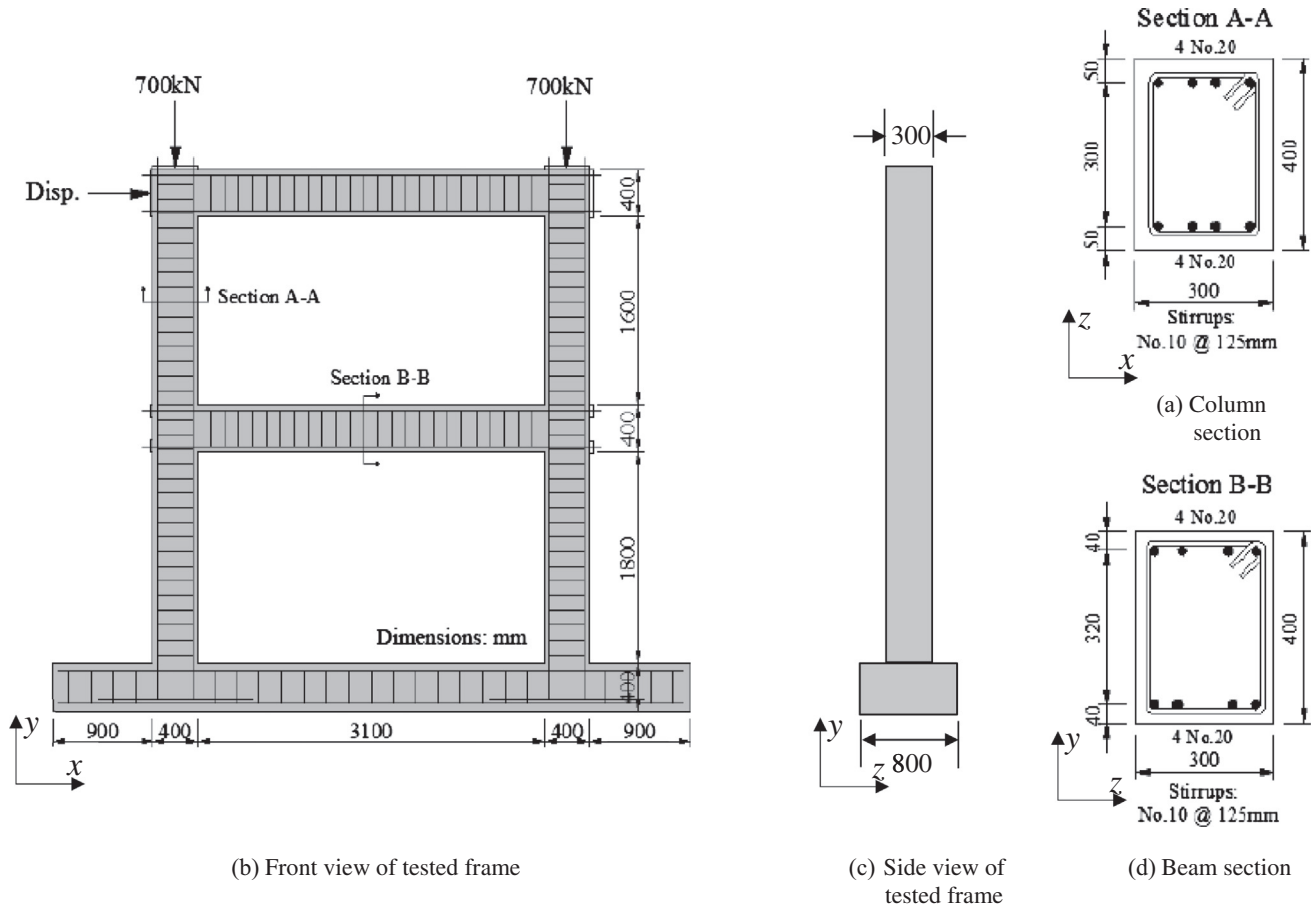
f_{ck} (MPa)	f_{cm} (MPa)	f_{tm} (MPa)	G_{ch} (N/mm)	G_F (N/mm)	b	a_c	a_t	l_{eq} (mm)	b_c	b_t
18	26	2.07	20.7	0.1312	0.8	7.873	1	125	310.48	2960

**Fig. 10.** Comparison between experimental and simulated results for the plain concrete test.

forces, and that pushing consisted in imposing a displacement law to the top left joint. Noticeably, since there were no distributed forces acting on the beams, there was no cracking prior to the lateral pushing. The tested frame was widely instrumented, thus providing extensive information of the damage progression.

Concrete mechanical parameters were determined from standard cylinder tests. The characteristic value of compressive strength is 30 MPa and the average value of secant deformation

modulus is 23.67 GPa. Poisson ratio and shear deformation modulus were estimated as 0.2 and 9.86 GPa, respectively. The steel parameters were obtained from coupon tests; the stress-strain plots were approximately trilinear: an initial linear elastic branch, a horizontal yielding plateau and a plastic hardening branch. Table 4 displays the most relevant figures regarding reinforcement steel. In Table 4, bars No. 20 and 10 correspond to longitudinal and transverse reinforcement, respectively (Fig. 11a and d). D_b is the

**Fig. 11.** First frame experiment [34].

bar diameter, f_y/f_u are the yield point/ultimate stress, E_s is the steel modulus of elasticity, E_{sh} is the slope of the hardening branch, ε_{sh} is the strain that corresponds to the onset of hardening, and ε_u is the ultimate strain.

Fig. 12 summarizes the results of the experiment. Fig. 12a displays plots of pushing force vs. top level displacement. The first observed damage was flexural cracking at the end sections of the first level beam, for force 52.5 kN and displacement 2 mm; this instant can be considered as the overall yield point, since, after that point, the capacity curve became nonlinear. Under force 145 kN and displacement 9.3 mm, flexural cracks were perceived at the columns bases, and shear cracks were simultaneously detected at the first story beam. When force reached 264 kN and displacement 26.4 mm, first steel yielding was perceived in the bottom tensioned longitudinal reinforcement of the left end section of the first story beam; for force 287 kN and displacement 31.6 mm, the top tensioned longitudinal reinforcement of the right end section of the first story beam also yielded. For force 323 kN and displacement 52.5 mm, the longitudinal reinforcements of the column bases yielded as well, and hinges at the ends of the first story beam failed; this failure involved yielding of longitudinal reinforcement, and crushing of compressed concrete. Then, for force 329 kN and displacement 74.7 mm, similar failure was apparent at the column bases. Almost simultaneously, same failure affected at ends of second story beam. Afterwards, lateral stiffness was almost non-existent; therefore, collapse mechanism consisted in formation of six hinges. The experiment was terminated, for pushing force 332 kN and lateral displacement 150 mm, due to stroke limitations of the actuator, see Fig. 12a. Fig. 12b and c display images of the damaged bottom section of the right column, and the top left connection, respectively.

The main objective of this experiment was to investigate the influence of shear-related effects in the overall structural behavior; the results showed that approximately 20% of the nonlinear lateral displacement was due to shear effects. Noticeably, at failure, around 12% of the overturning moment was due to $P-\Delta$ effects. Supplementary information regarding this experiment is available in [24].

This test had been previously simulated by some of the authors of this work [7,27,28], and by other researchers [34]. Barbat et al. used a viscous damage model, which was implemented in a fiber model with Timoshenko frame elements. Faleiro et al. implemented a damage plasticity formulation in a finite element model with planar 2D elements. Güner employed commercial software packages and an ad-hoc fiber model developed at University of Toronto.

The [79] test is simulated implementing the proposed methodology in Abaqus code [3] by using the particular algorithm described in Section 3.3. The maximum number of iterations is 10, and loading increment ranges between 10^{-11} mm and 0.01 mm. Table 5 displays the selected values of parameters. Fig. 13 displays the finite element mesh; the left part depicts steel discretization with 2-node truss elements (T3D2) and the right part describes concrete discretization with 3D 8-node hexahedron solid elements (C3D8R).

Fig. 14a displays plots of ratio $\varepsilon_c^{pl}/\varepsilon_c^{ch}$ vs. ε_c^{ch} ; Fig. 14b displays analogous plots for the tensile behavior. Fig. 14 highlights that, in the proposed methodology, damage and plastic energy absorp-

tions are not related, since ratios $\varepsilon_c^{pl}/\varepsilon_c^{ch}$ and $\varepsilon_t^{pl}/\varepsilon_t^{ch}$ do not approach zero when damage variables d_c and d_t are close to 1. Fig. 14a shows that the average value of ratio b (Eq. (28)) is approximately 0.75 (Table 5).

For comparison, simulations of [79] experiment with simplified concentrated and distributed plasticity models are also presented.

The concentrated plasticity model is implemented in software package SAP-2000 v.16 [23] by using the general frame formulation [8], without accounting for shear-generated deformation. Second-order effects are taken into account. Zero-length flexural and axial-flexural hinges are assigned to beams and columns, respectively. As well, shear hinges are considered for both types of members. Hinges are located at faces of joints. In flexural and axial-flexural hinges, parameters of initial and further branches are obtained according [4] and [31], respectively. In shear hinges, parameters of initial branch are obtained according to [4], and after-yield behavior is assumed to be totally brittle. Neither shear-flexural interaction nor shear-axial-flexural interaction is considered.

The distributed plasticity models are implemented in package SeismoStruct V6.5 [71]. Two types of models are utilized to simulate both frame experiments. First type adopts the classical displacement-based finite element formulation [39,55] and second type is based on the more recent force-based formulation [75,61]. Throughout this paper, these models are termed as DB and FB, respectively. FB approach does not impose any displacement field, and equilibrium is strictly and continuously satisfied. In the DB models, each member is discretized with four 2-node finite elements. In the FB models, a single element with five integration points represents each member. For both FB and DB models, sections are discretized into 250 fibers. Second-order effects are accounted for. As in the concentrated plasticity model, the interaction between shear and flexure is not taken into consideration.

Fig. 15 displays experimental results (Fig. 12a) plotted together with numerical results obtained with the proposed methodology and the abovementioned simplified models. Descriptions of observed damage states are also displayed.

Plots from Fig. 15 shows the superior ability of the proposed methodology to reproduce the experimental results along the whole displacement range. It captures the initial stiffness, the onset of overall yielding, the sequence of damage progression and the final state. Noticeably, “overall yielding” does not refer to steel yielding but to inception of overall nonlinear behavior, due to cracking of tensioned concrete. Regarding the lumped plasticity model, accuracy can be considered satisfactory, given the important simplifications involved in that model. Fig. 15 shows that the concentrated model describes satisfactorily the initial slope, but fails to predict the cracking and, therefore, the onset of overall yielding. The yielding branch is almost horizontal because of second order effects. The maximum capacity in terms of force is underestimated because of the conservative assumptions in the predefined plastic hinges. The final failure is earlier because the actual ductility of members is also underestimated by the assumed moment-rotation laws. Regarding the distributed plasticity models, Fig. 15 shows that they perform better, particularly FB. DB model exhibits less accuracy, because the mesh is too coarse. However, these models cannot capture adequately the gradual progression of the global softening after the initial overall yielding,

Table 4
Reinforcement steel parameters for the first frame experiment.

Bar No.	D_b (mm)	f_y (MPa)	f_u (MPa)	E_s (GPa)	E_{sh} (MPa)	ε_{sh}	ε_u
20	19.5	418	596	192.5	3100	0.0095	0.0669
10	11.3	454	640	200	3100	0.0095	0.0695

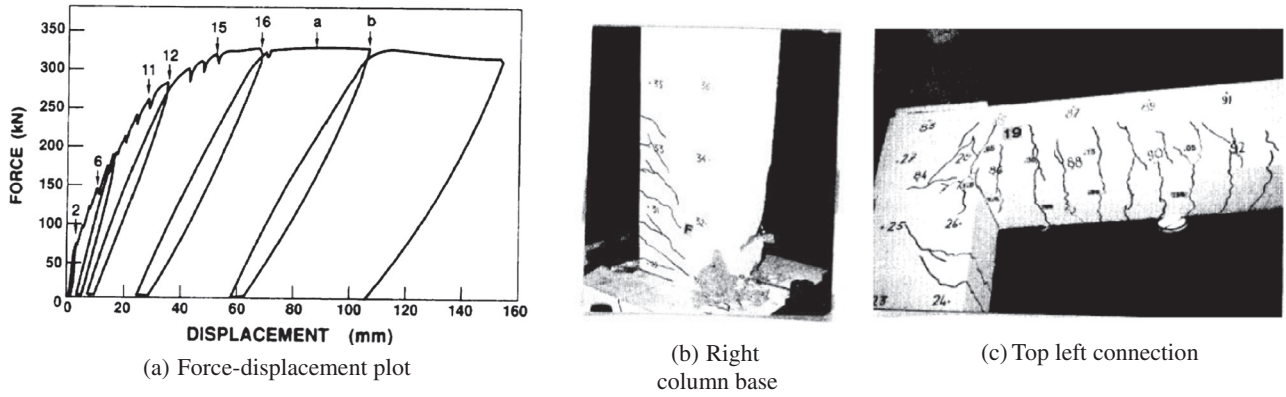


Fig. 12. Results of first frame experiment [79].

Table 5

Parameters for simulation of the first frame experiment.

f_{ck} (MPa)	f_{cm} (MPa)	f_{tm} (MPa)	G_{ch} (N/mm)	G_F (N/mm)	b	a_c	a_t	l_{eq} (mm)	b_c	b_t
30	38	2.912	23.93	0.1405	0.75	7.873	1	50	156.83	1554.54

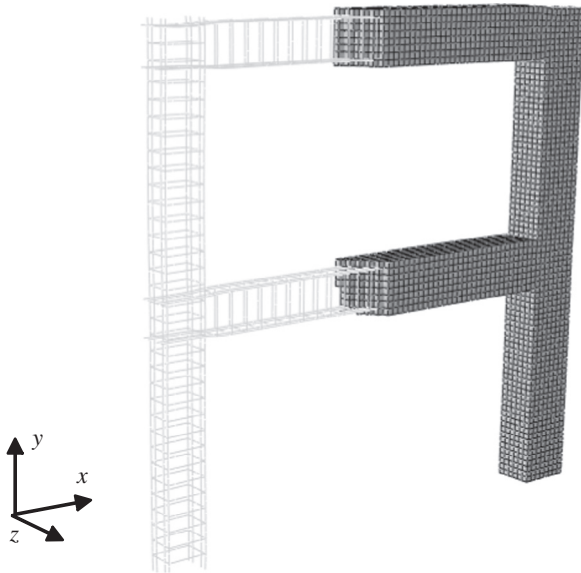


Fig. 13. Finite element discretization of the first frame experiment.

due to lack of consideration of the concrete tensile strength. Noticeably, the negative slope of the final branches is due to second-order effects.

To further highlight the capacity of the proposed methodology to capture damage progression, Figs. 16–18 display the damage predicted for some of the previously described stages. Fig. 16 displays the distribution of the tensile damage variable for force 52.5 kN and top level displacement 2 mm; it corresponds to the first cracking at end sections of 1st story beam. Fig. 16a and b refer to the first story right beam-column connection and to the overall frame, respectively. Fig. 16a shows that cracking (indicated with lighter gray) actually occurred in the top part of beam, since the tensile damage variable reaches values close to one. Fig. 16b shows that the overall distribution of cracking fits the expected pattern according to structural analysis principles, with onset of cracking in the bottom part of the first story right beam-column connection.

Figs. 17 and 18 refer to the final state. Fig. 17a and b display distributions of the scalar damage variable (d) for the right column base and the top left beam-column connection, respectively. Fig. 18 refers to the right column base, describing both the final state and the evolution from the undamaged state. Fig. 18a and c represent the distribution of the compressive and tensile damage variables, respectively; Fig. 18b and d display, for selected finite elements, plots of such variables vs. top level displacement, respec-

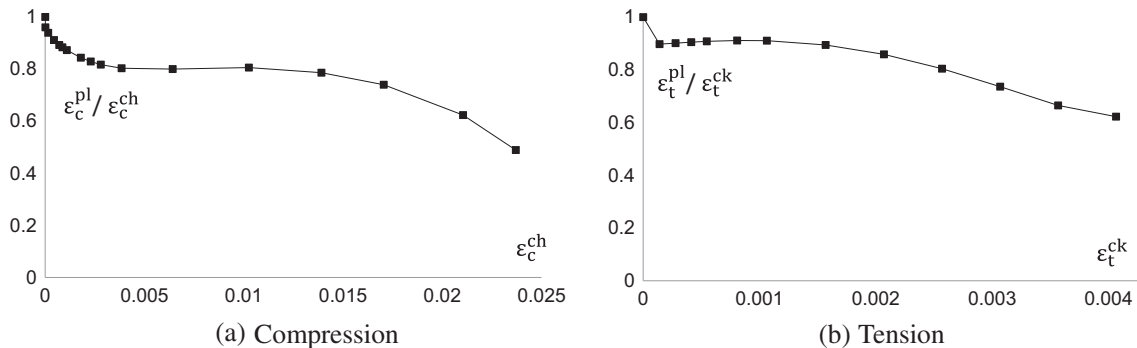


Fig. 14. Variation of ratio between plastic and crushing/cracking strains for the first frame experiment.

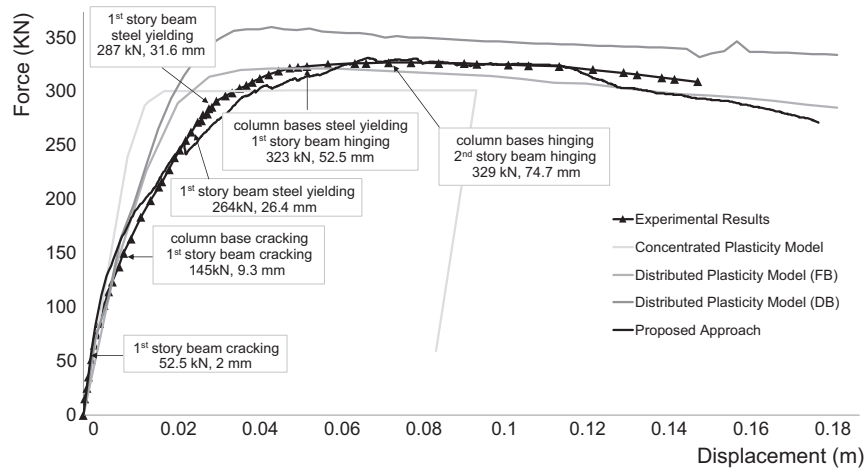


Fig. 15. Experimental and simulated capacity curves for the first frame experiment.

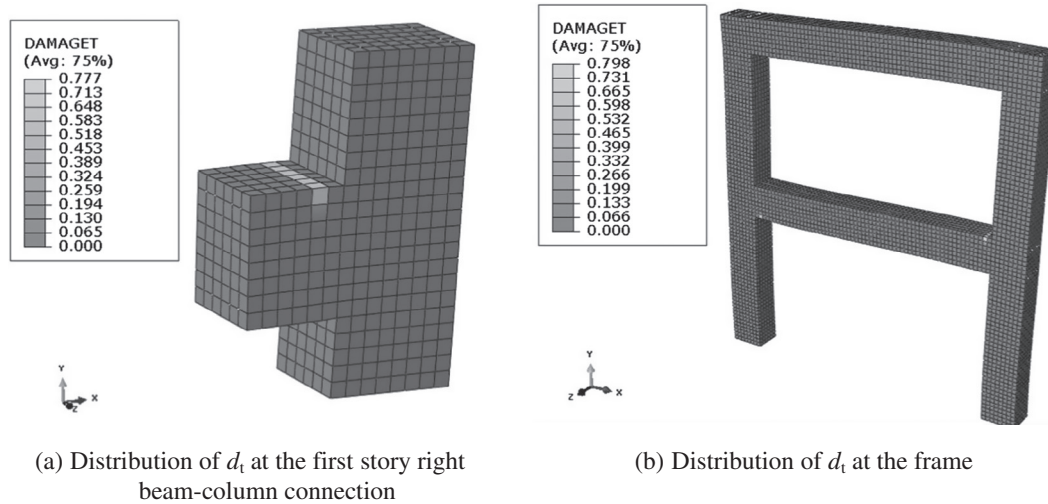


Fig. 16. Tensile damage variable for force 52.5 kN and displacement 2 mm for the first frame experiment.

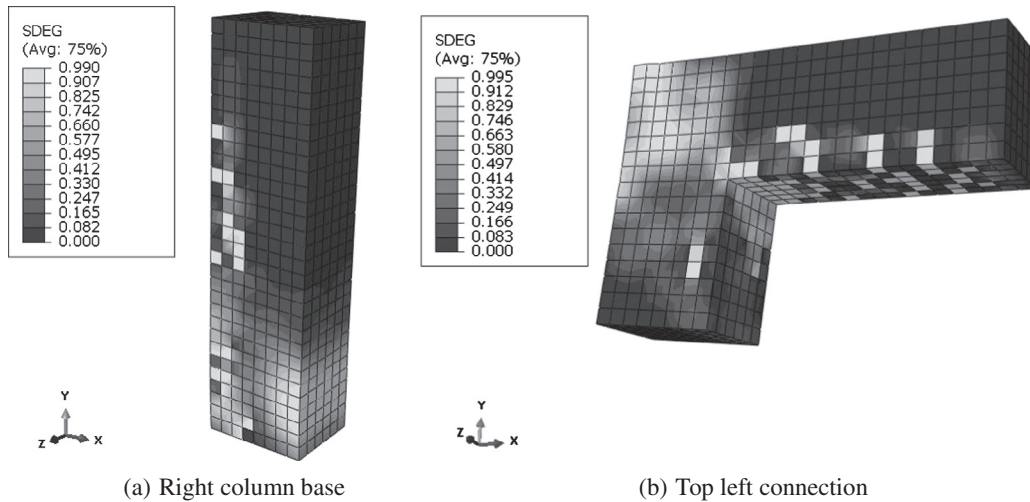


Fig. 17. Distribution of d at the final state of the first frame experiment.

tively. In all the images, higher damage corresponds to lighter grey. Figs. 17 and 18 show that the observed phenomena are adequately reproduced by the proposed methodology, since the obtained dam-

age distributions fit the expected results. Comparison between Fig. 17a and the observed damage in Fig. 12b shows a satisfactory fit, since cracking and crushing are detected by the obtained higher

values of d . Comparison between Fig. 17b and the observed damage in Fig. 12c shows also a satisfactory match. Fig. 18a and c shows that there is hinging, since both damage variables attain val-

ues close to 1; this circumstance is observed in Fig. 12b. Fig. 18b and d shows that cracking occurs for smaller displacement (approximately 10 mm) than crushing (more than 150 mm).

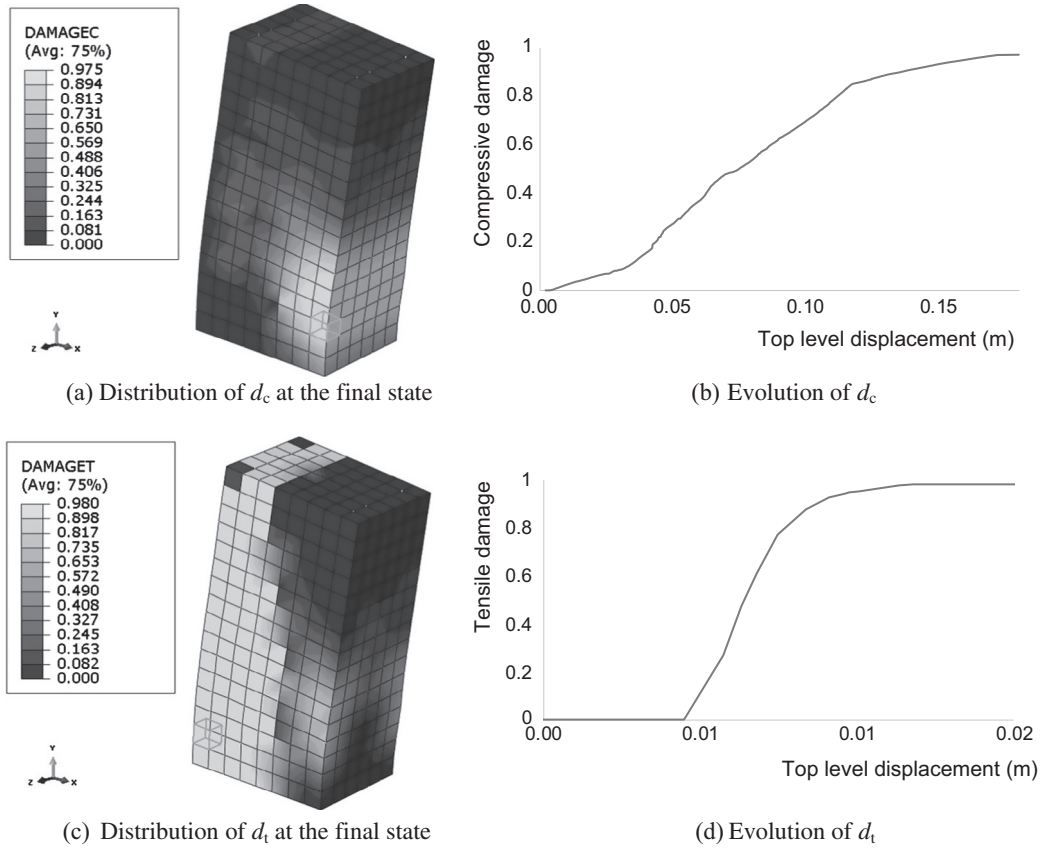


Fig. 18. Damage at the right column base for the first frame experiment.

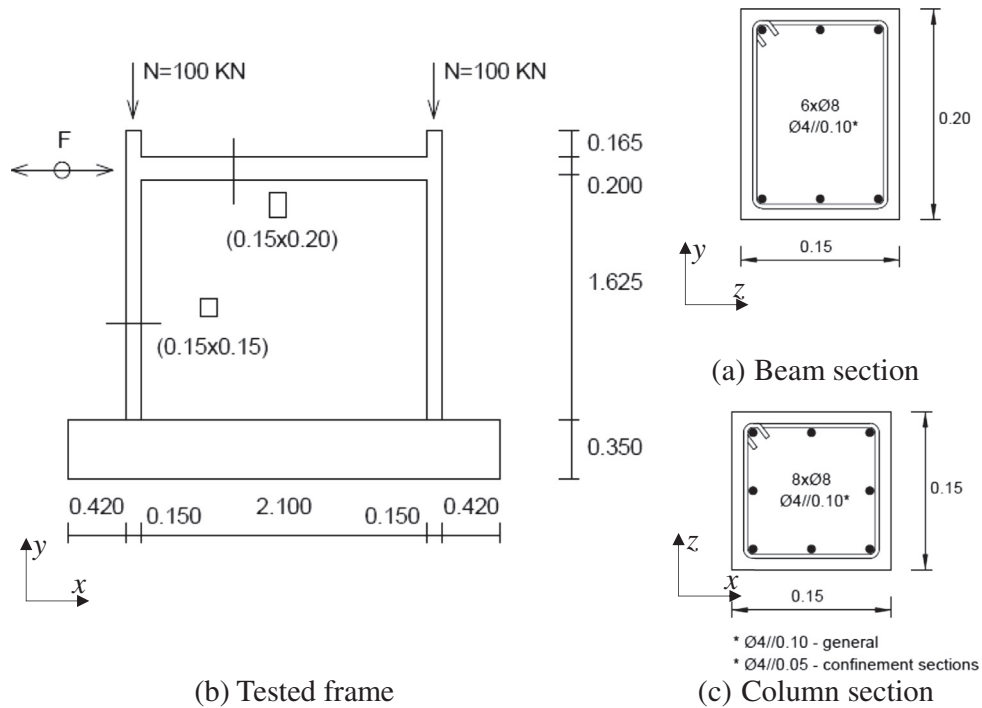


Fig. 19. Second frame experiment. [14,15]

4.2. Second frame test

This experiment [68] is a quasi-static test consisting of imposing a displacement law to a laboratory single-span, single-story 2-D RC frame. Fig. 19b describes the tested frame, and Fig. 19a and c display beam and columns sections, respectively. Fig. 19b shows that both columns were loaded with constant forces

and that displacement was imposed to the top left joint. Noticeably, as in the first frame test, given the absence of distributed loads on beams, there was no initial cracking.

Mechanical parameters of materials are based on nominal values. The characteristic value of the concrete compressive strength is 20 MPa (C20/25, [26]), and the steel yield point is 400 MPa for the longitudinal reinforcement and 500 MPa for the stirrups [25].

Table 6
Parameters for simulation of the second frame experiment.

f_{ck} (MPa)	f_{cm} (MPa)	f_{tm} (MPa)	G_{ch} (N/mm)	G_F (N/mm)	b	a_c	a_t	l_{eq} (mm)	b_c	b_t
20	28	2.222	21.12	0.133	0.9	7.873	1	25	65.48	626.67

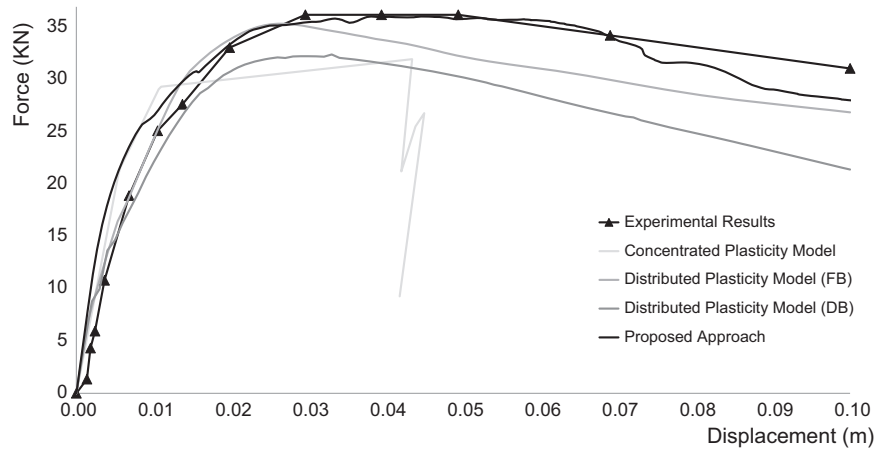


Fig. 20. Experimental and simulated capacity curves for the second frame experiment.

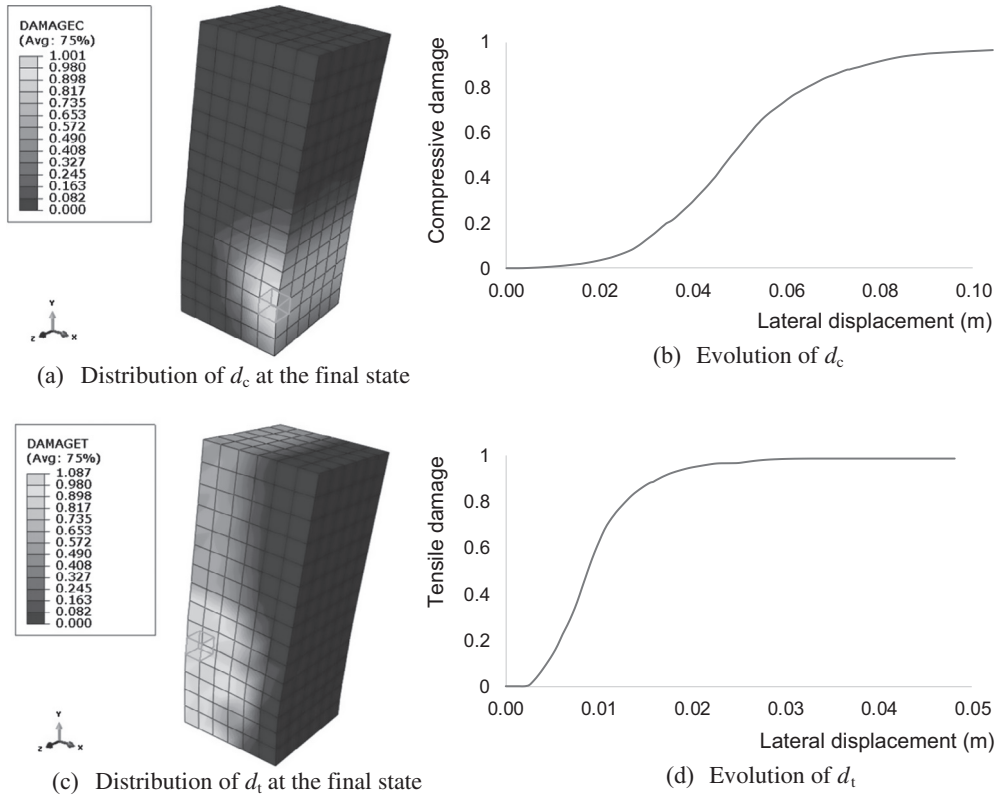


Fig. 21. Damage at the right column base for the second frame experiment.

As described in Fig. 19c, in the critical end segments (“confinement sections”), closer stirrup spacing was used; the lengths of these segments are 40 cm in beam and 30 cm in columns.

This frame had been previously simulated by [14,15] by using concentrated and distributed plasticity models.

Analogously to [79] experiment, this test is simulated implementing the algorithm in Abaqus code [3]. The maximum number of iterations is 10, and loading increment ranges between 0.0001 mm and 0.01 mm. Table 6 displays the selected values of the parameters.

The [68] experiment is simulated with the proposed methodology. For comparison, simulations with simplified concentrated and distributed plasticity models are also included. Similarly to Fig. 15, Fig. 20 displays experimental results together with numerical results from the proposed approach and the said simplified models. FB and DB refer to force-based and displacement-based, respectively.

Plots from Fig. 20 provide analogous conclusions than Fig. 15. The proposed methodology captures the initial stiffness, the inception of overall yielding, the sequence of damage progression and the final state. The concentrated plasticity model describes satisfactorily the initial slope, but fails to predict the cracking and, therefore, the onset of the overall yielding; force and displacement ductility are underestimated because of the conservative assumptions in the predefined plastic hinges. Distributed plasticity models perform better, particularly FB.

Fig. 21 shows analogous results than Fig. 18. Fig. 21 refers to the right column base, describing the final state and the evolution from the undamaged state. Fig. 21a and c represent the distribution of the compressive and tensile damage variables, respectively. Fig. 21b and d display, for selected finite elements, plots of such variables vs. beam displacement, respectively. Fig. 21a and c shows that there is hinging, since both damage variables attain values close to 1. Fig. 21b and d shows that cracking occurs for smaller displacement (approximately 20 mm) than crushing (more than 100 mm).

5. Conclusions

This paper presents a new methodology to calculate damage variables for Concrete Plastic Damage Models suitable for describing the monotonic behavior of reinforced concrete structures. The proposed approach is mesh-insensitive, is based on a sound continuum mechanics formulation, and does not require calibration with experimental results. Implementation is straightforward; a particular algorithm is presented. Mesh-insensitivity is validated in a simple tension example. Accuracy and reliability are verified by simulating a cyclic experiment on plain concrete specimens and two tests consisting in pushing monotonically until failure two laboratory 2-D RC frames. Results on these frames are compared with simplified models commonly employed in earthquake engineering.

Tension example. This example confirms the mesh-insensitiveness of the proposed methodology. As well, numerical results match analytical ones, thus ratifying their accuracy.

Cyclic experiment. The simulation of the cyclic experiment on plain concrete corroborates the capacity of the proposed approach to reproduce damage evolution and stress-strain behavior, both in terms of envelope curve and stiffness degradation at each unloading/reloading cycle. Agreement between experimental and numerical damage at measured strain levels is satisfactory.

Frame experiments. The simulation of the frame tests proves the ability of the proposed methodology to reproduce accurately the experimental force-displacement results along the whole displacement range. Captures initial stiffness, onset of the overall yielding, sequence of the damage progression, collapse mecha-

nism, and final state. All failure and degradation modes are adequately simulated: tensioned concrete cracking, compressed concrete crushing, and reinforcement steel yielding. The aforementioned simplified models provide less accuracy, being not able to reproduce all the involved phenomena.

Research for extending the proposed methodology to cyclic comportment of RC structures is currently in progress. The final objective is to derive models that are appropriate for simulating complex behavior of RC structures under strong ground motions.

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